Algoritmul QR - Metoda Householder

ortogonală

$A \in \mathbb{R}^{n \times n} \xrightarrow{\text{asemănare}} S = Q^T A Q = \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ & r_{22} & \dots & r_{2n} \\ & 0 & \ddots & \vdots \\ & & & r_{nn} \end{bmatrix}$

Forma Schur reală

$r_{ii} \in \begin{cases} \mathbb{R}^{1\times 1} \\ \mathbb{R}^{2\times 2} \end{cases}$ $\quad r_{ii} \begin{cases} \square \\ \boxplus \end{cases}$

ex $\begin{bmatrix} 1 & 0 & 2 & 3 & 1 & 4 \\ 0 & 2 & -1 & 3 & 0 & 1 \\ 0 & 1 & 2 & 3 & 0 & 1 \\ 0 & 0 & 0 & 7 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$

2 etape
1. directă
2. iterativă

$A \xrightarrow{①} \boxed{0} H \xrightarrow{②} S$

① aducerea lui A prin asemănare către forma superior Hessenberg

Pas 1 Reducerea la forma superior Hessenberg

Th $\forall A \in \mathbb{R}^{n\times n}$

$\exists U \in \mathbb{R}^{n\times n}$ ortogonală calculabilă printr-o procedură directă

astfel încât $H = U^T A U$

Construcția lui $U = U_2 \cdot U_3 \cdot \ldots \cdot U_{n-1}$

pentru $k = 1 : n-2$

- → se determină reflectorul $U_{k+1} \in \mathbb{R}^{n\times n}$ a.î. $(U_{k+1} A)_{(k+1:n,\,k)} = 0$
- → $A \leftarrow U_{k+1} A$
- → $A \leftarrow A U_{k+1}$

■

$$U_k = \mathcal{I}_n - \frac{V_k V_k^T}{P_k}$$

$k = 2 : n-1$

$$V_k(i) = \begin{cases} 0 & i = 1 : k-1 \\ \tau + x(k) & \\ x(i) & i = k+1 : n \end{cases}$$

unde $x = A(i, k)$

$P_k = V_k^T \cdot V_k$

$$\tau = \operatorname{sig}(x(k)) \cdot \|x(k:n)\|_2$$

mat sim → forma sup. Hessenb.
va fi tridiagonală

Pas 2 Pentru a ajunge la Forma Shur reală

$(H \longrightarrow S)$

→ $H_0 = H$

→ pentru $k = 0, 1, 2 \ldots$

→ $H_k - \mu_k \mathcal{I}_n = Q_k R_k$

→ $H_{k+1} = R_k Q_k + \mu_k \mathcal{I}_n$

Demo $H_{k+1} \sim H_k$

$Q_k^T \mid \cdot \; H_k - \mu_k \mathcal{I}_n = Q_k R_k \Rightarrow R_k = Q_k^T (H_k - \mu_k \mathcal{I}_n)$

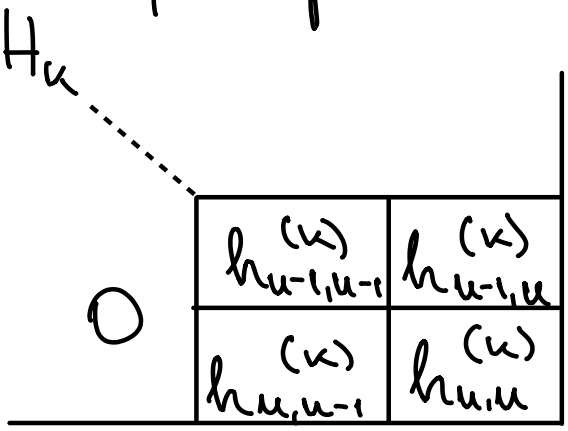
$H_{k+1} = Q_k^T (H_k - \mu_k \mathcal{I}_n) Q_k + \mu_k \mathcal{I}_n$

$= (Q_k^T H_k - \mu_k Q_k^T) Q_k + \mu_k \mathcal{I}_n$

$= Q_k^T H_k Q_k - \mu_k Q_k^T Q_k + \mu_k \mathcal{I}_n$

$= Q_k^T H_k Q_k \Rightarrow H_{k+1} \sim H_k$

Conditii oprire



$$\left| h^{(k+1)}_{k,k-1} - h^{(k)}_{k,k-1} \right| < \varepsilon$$

Alegerea lui $\mu_k = h^{(k)}_{k,k}$

$$\mu_k = h^{(k)}_{k-1,k-1} \cdot h^{(k)}_{k,k} - h^{(k)}_{k-1,k} \cdot h^{(k)}_{k,k-1}$$

Pasul $H_k - \mu_k \mathcal{I}_k = Q_k R_k$

→ $\mu = h_{k,k}$

→ $H \leftarrow H - \mu \mathcal{I}_k$

→ pentru $j = 1 : k-1$

→ se determină rotația $G_{j,j+1}$ a.î.

$$(G_{j,j+1} H)(j+1, j) = 0$$

→ $H \leftarrow G_{j,j+1} \cdot H$

→ pentru $j = 1 : k-1$

$H \leftarrow H \cdot G$

$G = \prod_j G_{j,j+1}$

! pentru asemănătoare

→ $H \leftarrow H + \mu \mathcal{I}_k$

Descompunerea valorilor singulare (SVD)

Th $\forall A \in \mathbb{R}^{m \times n}$

$\exists U, V \in \mathbb{R}^{m \times n}$ ortogonale astfel încât $A = U \Sigma V^T$

și $\Sigma = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}$ unde $\Sigma_r = \text{diag}(\sigma_1, \sigma_2 \dots \sigma_r)$

$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots \geq \sigma_r$

r = rangul matr A

$$B = A^T \cdot A = (U \Sigma V^T)^T \cdot (U \Sigma V^T) = V \Sigma^T U^T U \Sigma V^T = V \Sigma^2 V^T$$

(Σ, I_n)

$$\lambda(B) = \lambda(\Sigma^2) = \text{diag}(\sigma_1^2, \sigma_2^2 \dots \sigma_r^2, 0 \dots, 0)$$

$$\sigma_i = \sqrt{\lambda_i} \qquad \lambda_i \in \lambda(B)$$

$$C = A \cdot A^T = U \Sigma^2 U^T \Rightarrow \sigma_i = \sqrt{\lambda_i} \qquad \lambda_i \in \lambda(C)$$

$\text{rang}(A) = r$

$$\kappa(A) = \|A\| \cdot \|A^{-1}\| = \frac{\sigma_1}{\sigma_r}$$

κ = nr. de condiționare

$$A = U \Sigma V^T = \begin{bmatrix} U_1 \\ X \end{bmatrix} \begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V^T & X \end{bmatrix} = U_1 \Sigma_1 V_1^T$$

$$\Rightarrow A = \sum_{i=1}^{r} \sigma_i U_i V_i^T$$